# Chapter 10 exercises

State units, signs, and matrix shapes. Identify any missing boundary condition, normalization, or modulus before interpreting an application.

## CH10-E1

Consider graph with nodes $1, 2, 3, 4$ and oriented edges

$$1 \to 2, \quad 2 \to 3, \quad 1 \to 3, \quad 3 \to 4.$$

1. Write the $4 \times 4$ incidence matrix.
2. Determine connectivity, rank, and nullity.
3. Find a basis of $N(A^T)$ and identify its cycle.
4. Choose a node potential and a nonzero circulation, and verify $y^T Ax = 0$.

## CH10-E2

Give every edge in CH10-E1 unit conductance and remove edge $2 \to 3$, leaving a tree.

1. Write its edge-deleted incidence matrix $B$ and $L = B^T B$.
2. Check the Laplacian row and column sums.
3. Ground node 4 and give the reduced Laplacian.
4. Solve $L_{\text{red}}(x_1, x_2, x_3)^T = (1, 0, 0)^T$. Append $x_4 = 0$, compute the surviving edge differences $Bx$, and check the full node balance.

## CH10-E3

Three masses are connected by springs with stiffnesses $c_1, c_2, c_3, c_4$, fixed at both ends.

1. Give a $4 \times 3$ difference matrix satisfying $e = Au$.
2. Write $C$ and compute $K = A^T C A$.
3. For $c_i = 1$, find eigenvalues or prove positive definiteness.
4. Explain the nullspace and physical motion when both endpoints are free.

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## CH10-E4

Markov matrix

$$P = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}.$$

1. Check column sums and nonnegativity, and explain the column-stochastic convention.
2. Find eigenvalues and eigenvectors.
3. Normalize a stationary distribution.
4. For $u_0 = (1, 0)^T$, give $u_k$ and interpret the nonstationary mode.

## CH10-E5

Consumption matrix

$$A = \begin{bmatrix} 0.2 & 0.3 \\ 0.4 & 0.1 \end{bmatrix}.$$

1. Find $\rho(A)$.
2. Test invertibility and compute $(I - A)^{-1}$.
3. Find gross production for demand $y = (1, 2)^T$.
4. Explain how $\rho(A) < 1$ gives the Neumann series $I + A + A^2 + \cdots$.

## CH10-E6

Solve the linear program

$$\min 5x_1 + 3x_2 + 8x_3$$

subject to

$$x_1 + x_2 + 2x_3 = 4, \quad x_i \geq 0.$$

1. Describe the feasible region and its vertices.
2. Evaluate the objective at every vertex and find the optimum.
3. Write the dual and certify the gap.
4. Decide whether this problem has distinct optimal vertices. Explain how nonunique optimal production in a general LP differs from multiple bases at the same degenerate vertex.

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## CH10-E7

On $[0, 2\pi]$, let

$$f(x) = \begin{cases} 1, & 0 \le x < \pi, \\ -1, & \pi \le x \le 2\pi. \end{cases}$$

1. Compute the constant and cosine coefficients and explain the symmetry.
2. Compute sine coefficients for odd and even indices.
3. Write the finite truncation.
4. Interpret projection and describe the change in squared $L^2$ error.

## CH10-E8

Use homogeneous column coordinates.

1. Write homogeneous matrices for translation by $t = (2, -1)^T$, scaling by 3, and rotation by $90^\circ$.
2. Compose rotation about center $(2, -1)^T$.
3. Apply translation to point $p = (1, 0)^T$, represented homogeneously with final coordinate 1, and direction $d = (1, 0, 0)^T$. Explain the difference.
4. Give projection and reflection matrices for unit normal $n = (1, 2)^T/\sqrt{5}$, stating multiplication order.

## CH10-E9

Hill cipher over $\mathbb{F}_{29}$:

$$K = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}.$$

1. Compute the determinant modulo 29 and test invertibility.
2. Compute the modular inverse.
3. Encode $x = (5, 8)^T$.
4. Decode and explain why known plaintext can reveal the key.

## CH10-E10

Combine the network constructions.

Ground node 1 of a four-node supply network. The ordered edges are

$$1 \to 2, \qquad 1 \to 3, \qquad 2 \to 4, \qquad 3 \to 4,$$

with conductances $c = (1, 2, 3, 4)$ and injections, positive for sources and negative for sinks,

$$f = (-4, -4, -10, 18)^T.$$

1. Use head minus tail to construct $A$ and $L = A^T C A$. State the units and sign of every $f_i$.
2. Reverse the second edge, writing $A' = DA$. Find $D$, prove $L' = L$, and describe the consistent sign changes of differences and flows for the same physical state.
3. Check $\mathbf{1}^T f = 0$. Delete node 1's row and column, solve the reduced Laplacian, and reconstruct $x = (0, x_2, x_3, x_4)^T$.
4. Compute $e = Ax$, $y = Ce$, and physical currents $j = -y$. Verify $-A^T j = A^T y = f$ and $\frac{1}{2} x^T L x = \frac{1}{2} \sum_e c_e e_e^2$.
5. If the last source is mistakenly recorded as $17$, why is $Lx = f$ inconsistent? Can a grounded solve conceal this error, and how should it be corrected?

## CH10-E11

Minimize $-x_1$ with $x_1 + s_1 = 0$, $x_1 + s_2 = 1$, and all variables nonnegative.

1. Starting with basis $s_1, s_2$, give the initial point, entering variable, and every ratio. Perform one valid pivot.
2. If only strictly positive ratios are allowed, give the resulting infeasible point and identify the violated condition.
3. Prove from the new dictionary that the optimum is 0. Explain why a basis change need not change the vertex or objective.
4. For standard-form $\min c^T x$, $Ax = b$, $x \geq 0$, assuming $A$ has full row rank and the current basic solution is feasible, prove that nonnegative reduced costs certify global optimality. Explain why a negative reduced cost and nonpositive direction denominators certify unboundedness.

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## CH10-E12

For the weighted triangle with ordered edges $1 \to 2, 2 \to 3, 1 \to 3$ and conductances $(2, 3, 1)$ S, take node 1 as the potential reference. Solve the 2-by-2 reduced system with right-hand side $(-3,15)^T$, then recover the omitted injection $f_1$. Check physical current directions, total injection and dissipated power when potentials are in volts and conductances in siemens. If $f_1$ is recorded as −11 A, why can the reduced solve still succeed while the complete system is inconsistent? Does $\frac{1}{2}x^TLx$ represent energy stored in ideal resistors?

### References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 10.
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